

Week 3 Progress Report

1. Objectives for Week 3:

- Implement preprocessing techniques for X-ray and CT scan datasets.
- Apply image enhancement (noise reduction, normalization, resizing).
- Begin feature extraction using CNN-based architectures.

2. Activities Completed:

1. Data Preprocessing:

- Applied **image resizing** (224x224 px) for model compatibility.
- Implemented **normalization** to standardize pixel intensity values.
- Used **histogram equalization** to improve image clarity.

2. Data Augmentation:

- Applied transformations: rotation, flipping, scaling, and zooming to balance the dataset.
- Increased dataset diversity by ~30% to avoid overfitting.

3. Feature Extraction Setup:

- Configured CNN architecture (baseline: VGG16, ResNet-50).
- Extracted low-level features such as edges, textures, and lung region patterns.

3. Challenges Faced:

- High computation time during preprocessing and augmentation.
- Class imbalance between cancerous and non-cancerous images.

4. Outcomes of Week 3:

- Created a **clean and preprocessed dataset** ready for training.
- Improved image quality with augmentation and normalization.
- Extracted **initial features using CNN**, confirming suitability for classification.

Week 4 Progress Report

1. Objectives for Week 4:

- Train initial machine learning and deep learning models.
- Compare performance of classical ML (SVM, Random Forest) vs. DL (CNN).
- Evaluate models using accuracy, precision, recall, and F1-score.

2. Activities Completed:

1. Model Training:

- Trained **CNN (ResNet-50, VGG16)** on preprocessed dataset.
- Implemented **SVM and Random Forest** using extracted CNN features.

2. Model Evaluation:

- Used **80:20 split** for training and testing.
- Evaluated models on metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC.
- ResNet-50 achieved **~85% accuracy**, outperforming SVM (~74%) and Random Forest (~70%).

3. Hyperparameter Tuning:

- Optimized batch size, learning rate, and number of epochs.
- Applied dropout regularization to reduce overfitting.

3. Challenges Faced:

- Longer training time for deep CNN models.
- Overfitting is observed in smaller datasets without augmentation.

4. Outcomes of Week 4:

- Deep learning (ResNet-50, VGG16) showed **better accuracy** than classical ML methods.
- Identified **ResNet-50 as the baseline model** for further optimization.
- Prepared comparative results for the Week 5 model improvement phase.